

Actividades Unidad 4 Métodos Numéricos

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4.1 Diferenciación numérica - E1

$$f(x) = x^2$$

Calcular $f'(2)$ con $h=0.1$

usar: adelante, atrás y centrada

$$f(2) = 2^2 = 4$$

$$f(2+0.1) = f(2.1) = 2.1^2 = 4.41$$

$$f(2-0.1) = f(1.9) = 1.9^2 = 3.61$$

Fórmula hacia adelante

$$f'(2) = [f(2.1) - f(2)] / 0.1$$

$$f'(2) = (4.41 - 4) / 0.1$$

$$f'(2) = 0.41 / 0.1$$

$$f'(2) = 4.1$$

Fórmula hacia atrás

$$f'(2) = [f(2) - f(1.9)] / 0.1$$

$$f'(2) = (4 - 3.61) / 0.1$$

$$f'(2) = 0.39 / 0.1$$

$$f'(2) = 3.9$$

Fórmula centrada

$$f'(2) = [f(2.1) - f(1.9)] / 2(0.1)$$

$$f'(2) = (4.41 - 3.61) / 0.2$$

$$f'(2) = 0.8 / 0.2$$

$$f'(2) = 4.0$$

E 2 Ejercicio

$F(x) = \sin(x)$, Calcular $F'(\pi/4)$ con $h=0.05$

Usar fórmula centrada.

- Fórmula

$$F'(x) = [F(x+h) - F(x-h)] / 2h$$

- Datos:

$$x = \pi/4 = 0.7854$$

$$h = 0.05$$

$$x+h = 0.8354$$

$$x-h = 0.7354$$

- Sustitución

$$F'(\pi/4) = [\sin(0.8354) - \sin(0.7354)] / 2(0.05)$$

$$F'(\pi/4) = [0.7416 - 0.6710] / 0.1$$

$$F'(\pi/4) = 0.0707 / 0.1$$

$$F'(\pi/4) = 0.7071$$

E3

$f(x) = e^x$, calcular $f'(1)$ con $h = 0.1$.
Comparar los 3 métodos

• Datos

$$f(x) = e^x$$

$$x = 1$$

$$h = 0.1$$

• Valores

$$f(1) = e^1 = 2.7183$$

$$f(1.1) = e^{1.1} = 3.0042$$

$$f(0.9) = e^{0.9} = 2.4596$$

• Adelante

$$f'(1) = [f(1.1) - f(1)] / 0.1$$

$$f'(1) = (3.0042 - 2.7183) / 0.1$$

$$f'(1) = 2.859$$

• Atrás

$$f'(1) = [f(1) - f(0.9)] / 0.1$$

$$f'(1) = (2.7183 - 2.4596) / 0.1$$

$$f'(1) = 2.587$$

• Centrada

$$f'(1) = [f(1.1) - f(0.9)] / 0.2$$

$$f'(1) = (3.0042 - 2.4596) / 0.2$$

$$f'(1) = 2.723$$

• Valor exacto

$$f'(x) = e^x$$

$$f'(1) = e^1 = 2.7183$$

La fórmula centrada es más precisa

E4

$f(x) = \ln(x)$, calcular $f'(2)$ con $h = 0.1$.

Calcular error vs. exacto

• Fórmula centrada

$$f'(x) = [f(x+h) - f(x-h)] / 2h$$

• Datos

$$x = 2$$

$$h = 0.1$$

$$x+h = 2.1$$

$$x-h = 1.9$$

• Valores

$$\ln(2.1) = 0.7419$$

$$\ln(1.9) = 0.6419$$

• Sustitución

$$f'(2) = [\ln(2.1) - \ln(1.9)] / 0.2$$

$$f'(2) = (0.7419 - 0.6419) / 0.2$$

$$f'(2) = 0.1000 / 0.2$$

$$f'(2) = 0.5$$

• Valor exacto

$$f'(x) = 1/x$$

$$f'(2) = 1/2 = 0.5$$

• Error

$$\text{Error} = |\text{valor exacto} - \text{valor aproximado}|$$

$$\text{Error} = |0.5 - 0.5|$$

$$\text{Error} = 0.01$$

$$\text{Valor exacto} = 0.5$$

$$\text{Error aproximado} = 0.01$$

E5

$f(x) = x^3$, calcular $f''(1)$ con $h = 0.1$

• Formula

$$f''(x) = [f(x+h) - 2f(x) + f(x-h)] / h^2$$

• Datos

$$f(x) = x^3$$

$$x = 1$$

$$h = 0.1$$

• Valores

$$f(1.1) = 1.1^3 = 1.331$$

$$f(1) = 1^3 = 1$$

$$f(0.9) = 0.9^3 = 0.729$$

• Sustitución

$$f''(1) = [f(1.1) - 2f(1) + f(0.9)] / (0.1)^2$$

$$f''(1) = [1.331 - 2(1) + 0.729] / 0.01$$

$$f''(1) = [1.331 - 2 + 0.729] / 0.01$$

$$f''(1) = 0.060 / 0.01$$

$$f''(1) = 6.0$$

$$\underline{f''(1) = 6.0}$$

4.2 Regla del Trapecio E6

$\int_0^1 x^2 dx$ con $n=4$

• Datos

$$a=0$$

$$b=1$$

$$n=4$$

$$f(x)=x^2$$

• calcular h

$$h=(1-0)/4$$

$$h=0.25$$

• calcular valores x

$$x_0=0$$

$$x_1=0.25$$

$$x_2=0.50$$

$$x_3=0.75$$

$$x_4=1$$

• Evaluar la función

$$f(0)=0^2=0$$

$$f(0.25)=0.25^2=0.0625$$

$$f(0.50)=0.50^2=0.25$$

$$f(0.75)=0.75^2=0.5625$$

$$f(1)=1^2=1$$

• Sustituir

$$I=(0.25/2)[0+2(0.0625)+2(0.25)+2(0.5625)+1]$$

$$I=0.125[0+0.125+0.5+1.125+1]$$

$$I=0.125(2.75)$$

$$I=0.34375$$

$$I=0.3438$$

Ej $\int_0^{\pi} \sin(x) dx$ con $n=4$

• Datos

$$a=0$$

$$b=\pi$$

$$n=4$$

$$f(x) = \sin(x)$$

• Calcular h

$$h = (\pi - 0) / 4$$

$$h = \pi / 4$$

$$h = 0.7854$$

• Calcular valores de x

$$x_0 = 0$$

$$x_1 = \pi / 4$$

$$x_2 = \pi / 2$$

$$x_3 = 3\pi / 4$$

$$x_4 = \pi$$

• Evaluar la función

$$f(0) = \sin(0) = 0$$

$$f(\pi/4) = \sin(\pi/4) = 0.7071$$

$$f(\pi/2) = \sin(\pi/2) = 1$$

$$f(3\pi/4) = \sin(3\pi/4) = 0.7071$$

$$f(\pi) = \sin(\pi) = 0$$

• Sumar

$$I = (0.7854/2) [0 + 2(0.7071) + 2(1) + 2(0.7071) + 0]$$

$$I = 0.3927 [1.4142 + 2 + 1.4142]$$

$$I = 0.3927 (4.8284)$$

$$I = 1.8961$$

E8 $\int_1^2 1/x \, dx$ con $n=4$ (calcular $\ln(2)$)

• Datos

$a=1$

$b=2$

$n=4$

$f(x) = 1/x$

• Calcular h

$h = (2-1)/4$

$h = 0.25$

• Calcular valores de x

$x_0 = 1$

$x_1 = 1.25$

$x_2 = 1.50$

$x_3 = 1.75$

$x_4 = 2$

• Evaluar la función

$f(1) = 1/1 = 1$

$f(1.25) = 1/1.25 = 0.8$

$f(1.50) = 1/1.50 = 0.6667$

$f(1.75) = 1/1.75 = 0.5714$

$f(2) = 1/2 = 0.5$

• Sustituir

$I = (0.25/2) [1 + 2(0.8) + 2(0.6667) + 2(0.5714) + 0.5]$

$I = 0.125 [1 + 1.6 + 1.3334 + 1.1428 + 0.5]$

$I = 0.125 (5.5762)$

$I = 0.6970$

$I = 0.6970$

Eq $\int_0^1 e^x dx$ con $n=6$

• Datos

$a=0$

$b=1$

$n=6$

$f(x)=e^x$

• Calcular h

$h=(1-0)/6$

$h=0.1667$

• Calcular valores x

$x_0=0$

$x_1=0.1667$

$x_2=0.3333$

$x_3=0.5000$

$x_4=0.6667$

$x_5=0.8333$

$x_6=1$

• Evaluar la función

$f(0)=e^0=1$

$f(0.1667)=1.1814$

$f(0.3333)=1.3956$

$f(0.5000)=1.6487$

$f(0.6667)=1.9477$

$f(0.8333)=2.3010$

$f(1)=e^1=2.7183$

• Sustituir

$$I = (0.1667/2) [1 + 2(1.1814) + 2(1.3956) + 2(1.6487) + 2(1.9477) + 2(2.3010) + 2.7183]$$

$$I = 0.08335 [1 + 2.3628 + 2.7912 + 3.2974 + 3.8954 + 4.6020 + 2.7183]$$

$$I = 0.08335 (20.6671)$$

$I = 1.7223$

4.3 Regla de Simpson E10

So $x^2 dx$ con simpson $1/3, n=4$

- Fórmula general

$$h = (b-a) / n$$

$$I = (h/3) [f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \dots + f(x_n)]$$

- Datos

$$a = 0$$

$$b = 1$$

$$n = 4$$

$$f(x) = x^2$$

- Calcular h

$$h = (1-0) / 4$$

$$h = 0.25$$

- Valores

$$x_0 = 0 \quad f(x_0) = 0$$

$$x_1 = 0.25 \quad f(x_1) = 0.0625$$

$$x_2 = 0.50 \quad f(x_2) = 0.25$$

$$x_3 = 0.75 \quad f(x_3) = 0.5625$$

$$x_4 = 1 \quad f(x_4) = 1$$

- Sustitución

$$I = (0.25/3) [0 + 4(0.0625) + 2(0.25) + 4(0.5625) + 1]$$

$$I = 0.0833 [0 + 0.25 + 0.5 + 2.25 + 1]$$

$$I = 0.0833 (4)$$

$$I = 0.3333$$

$$I = 0.3333$$

E11 $\int_0^\pi \sin(x) dx$ con $n=4$

• Datos

$$a=0$$

$$b=\pi$$

$$n=4$$

$$f(x) = \sin(x)$$

• Calcular h :

$$h = (\pi - 0) / 4$$

$$h = \pi / 4$$

$$h = 0.7854$$

• Valores

$$x_0 = 0 \quad f(x_0) = 0$$

$$x_1 = \pi/4 \quad f(x_1) = 0.7071$$

$$x_2 = \pi/2 \quad f(x_2) = 1$$

$$x_3 = 3\pi/4 \quad f(x_3) = 0.7071$$

$$x_4 = \pi \quad f(x_4) = 0$$

• Substitution

$$I = (0.7854/3) [0 + 4(0.7071) + 2(1) + 4(0.7071) + 0]$$

$$I = 0.2618 [2.8284 + 2 + 2.8284]$$

$$I = 0.2618 (7.6568)$$

$$I = 2.0046$$

$$I = 2.0046$$

E 12 $\int_0^2 x^3 dx$ con $n=6$ con $x_6 (x) = 2 \pi 0$

• Datos

$$a = 0$$

$$b = 2$$

$$n = 6$$

$$f(x) = x^3$$

• calcular h :

$$h = (2-0) / 6$$

$$h = 0.3333$$

• Valores

$$x_0 = 0 \quad f(x_0) = 0$$

$$x_1 = 0.3333 \quad f(x_1) = 0.0370$$

$$x_2 = 0.6667 \quad f(x_2) = 0.2963$$

$$x_3 = 1.0000 \quad f(x_3) = 1$$

$$x_4 = 1.3333 \quad f(x_4) = 2.3704$$

$$x_5 = 1.6667 \quad f(x_5) = 4.6296$$

$$x_6 = 2 \quad f(x_6) = 8$$

• Sustitución

$$I = (0.3333/3) [0 + 4(0.0370) + 2(0.2963) + 4(1) + 2(2.3704) + 4(4.6296) + 8]$$

$$I = 0.1111 [0 + 0.148 + 0.5926 + 4 + 4.7408 + 18.5184 + 8]$$

$$I = 0.1111 (36)$$

$$I = 4.0$$

E13 Si $\int_1^3 \frac{1}{x} dx$ con $n=4$

• Datos

$$a=1$$

$$b=3$$

$$n=4$$

$$f(x) = 1/x$$

• Calcular h

$$h = (3-1)/4$$

$$h = 0.5$$

• Valores

$$x_0 = 1$$

$$f(x_0) = 1$$

$$x_1 = 1.5$$

$$f(x_1) = 0.6667$$

$$x_2 = 2$$

$$f(x_2) = 0.5$$

$$x_3 = 2.5$$

$$f(x_3) = 0.4$$

$$x_4 = 3$$

$$f(x_4) = 0.3333$$

• Sustitución

$$I = (0.5/3) [1 + 4(0.6667) + 2(0.5) + 4(0.4) + 0.3333]$$

$$I = 0.1667 [1 + 2.6668 + 1 + 1.6 + 0.3333]$$

$$I = 0.1667 (6.6001)$$

$$I = 1.1000$$

$$I = 1.0986$$

$$\text{Exacto} = 1.099$$

E 14 $\int_0^1 e^{-x^2} dx$ con $n=6$

• Datos

$a=0$

$b=1$

$n=6$

$f(x)=e^{-x^2}$

• Calcular h

$h=(1-0)/6$

$h=0.1667$

• Valores aproximados

$x_0=0 \quad f(x_0)=1$

$x_1=0.1667 \quad f(x_1)=0.9726$

$x_2=0.3333 \quad f(x_2)=0.8948$

$x_3=0.5000 \quad f(x_3)=0.7788$

$x_4=0.6667 \quad f(x_4)=0.6412$

$x_5=0.8333 \quad f(x_5)=0.4994$

$x_6=1 \quad f(x_6)=0.3679$

• Sustitución

$$I = (0.1667/3) [1 + 4(0.9726) + 2(0.8948) + 4(0.7788) + 2(0.6412) + 4(0.4994) + 0.3679]$$

$$I = 0.0556 [1 + 3.8904 + 1.7896 + 3.1152 + 1.2824 + 1.9976 + 0.3679]$$

$$I = 0.0556 (13.4431)$$

$$I = 0.7468$$

$$I = 0.7468$$

4.4 Cuadratura Gaussiana
E15 $\int_{-1}^1 x^2 dx$ con 2 puntos Gauss

• Datos

$$f(x) = x^2$$

$$x_1 = -0.5774$$

$$x_2 = 0.5774$$

• Evaluar la función

$$f(-0.5774) = (-0.5774)^2$$

$$f(-0.5774) = 0.3333$$

$$f(0.5774) = (0.5774)^2$$

$$f(0.5774) = 0.3333$$

• Sustitución

$$I = f(-0.5774) + f(0.5774)$$

$$I = 0.3333 + 0.3333$$

$$I = 0.6666$$

$$\underline{I = 0.6667}$$

E16 $\int_0^1 e^x dx$ transformando a $[-1, 1]$

• Datos

$$a = 0$$

$$b = 1$$

$$f(x) = e^x$$

• Fórmula de Transformación
 $x = ((b-a)/2)t + ((b+a)/2)$

• Sustitución

$$x = ((1-0)/2)t + ((1+0)/2)$$

$$x = 0.5t + 0.5$$

$$dx = ((b-a)/2) dt$$

$$dx = 0.5 dt$$

• Puntos de Gauss

$$t_1 = -0.5774$$

$$t_2 = 0.5774$$

• Transformar

$$x_1 = 0.5(-0.5774) + 0.5$$

$$x_1 = -0.2887 + 0.5$$

$$x_1 = 0.2113$$

$$x_2 = 0.5(0.5774) + 0.5$$

$$x_2 = 0.2887 + 0.5$$

$$x_2 = 0.7887$$

• Evaluar la función

$$f(x_1) = e^{0.2113} = 1.2353$$

$$f(x_2) = e^{0.7887} = 2.2003$$

• Aplicar fórmula

$$I = ((b-a)/2) [f(x_1) + f(x_2)]$$

$$I = 0.5 [1.2353 + 2.2003]$$

$$I = 0.5 (3.4356)$$

$$I = 1.7178$$

$$I = 1.7178$$